MIL-L-51543(EA)
12 July 1988
(See section 6)

MILITARY SPECIFICATION

LAUNCHER, PROJECTILE, LIQUID AIRBURST: M267

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid airburst projectile launcher.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood

MILITARY

MIL-C-49104 - Cable, Telephone WD-1A/TT (Field Wire, Parallel Pair)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter).

AMSC N/A FSC 6920

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TDPL 13-12-145 - Launcher, Projectile, Liquid Airburst, M267

PUBLICATIONS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TM 1370-102-12 - Operator's and Organizational Maintenance Manual, Simulator, Projectile, Airburst Liquid: M11 Launcher, Projectile, Liquid Airburst: M267

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM STANDARDS

D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on drawings in TDPL 13-12-145 and on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the launcher shall conform to the specifications and drawings listed in TDPL 13-12-145 and on subsidiary drawings.

3.1.2.1 Molded plastic parts. The launcher tube shall be manufactured to conform to the following properties:

(a) Color. Color shall be uniform prior to applying surface finish.

(b) Surface. The molded component surfaces shall be free of tool marks and possess a smooth finish.

(c) Cracks. Molded parts shall be free from porosity, warpage, cracks, chipped edges, scratches, and other defects affecting serviceability.

(d) Markings. Each molded part shall have molded thereon the manufacturer's trademark, cavity number, and molding date marks as required by the applicable drawings, specifications, contract or purchase order.

3.2 Manufacture and assembly. The launcher tubes shall be produced and assembled in accordance with 13-12-145.

3.3 Functioning.

3.3.1 Burst strength. The launcher tube (Drawing 13-12-141) shall be capable of withstanding a minimum hydrostatic pressure of 450 pounds per square inch gage (psig) for a minimum time of 2 minutes without bursting or cracking when tested as specified in 4.4.4.1.

3.3.2 Electrical connector (Drawing 13-12-146).

3.3.2.1 Electrical short. The electrical connector test clips shall not be electrically shorted out on the metal box they are mounted on when tested as specified in 4.4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

3.3.2.2 Electrical continuity. The electrical connector test clips for igniter wires and corresponding test clips for firing line shall possess electrical continuity when tested as specified in 4.4.4.2.

3.3.3 Continuity check of electrical telephone cable. The electrical continuity of the telephone cable shall follow procedures outlined in MIL-C-49104.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of launchers shall be produced in accordance with this specification for examinations and tests.

3.5 Workmanship. The launcher and its components shall be free from imperfection, damage (bent, warped, cracked, flash-burned metal sections), and foreign matter (oil, grease, or dirt).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of components suitable to assemble 50 launchers shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Examination. A sample of components, subassemblies, and assemblies for 10 launchers and two unit packed and packed launchers shall be examined for all requirements of the drawings and of this specification. Assemble 20 launchers without any difficulties.

4.3.3 Destructive test. Ten launcher tubes shall be burst tested as specified in 4.4.4.1.

4.3.4 Acceptance/rejection criteria. The preproduction samples shall meet the examination and tests specified in 4.3.2 and 4.3.3 to be acceptable. The acceptance number is zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of no more than 2,000 launchers using like components or subassemblies of one interfix number from one manufacturer, from the same materials, and under essentially the same manufacturing conditions. A lot of launcher tubes shall consist of no more than 10,000 launcher tubes.

4.4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure for examination and tests. Samples shall be examined and tested in accordance with classification of defects and with MIL-STD-105.

4.4.3.1 For destructive tests. Sampling for the burst strength test shall be conducted in accordance with MIL-STD-105, level S-3, using an AQL of 0.15 percent defective.

4.4.3.2 Classification of defects.

(a) Launcher, projectile, liquid airburst: M267 (Drawing 13-12-145).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Components missing
102	Workmanship (3.5)

Source: https://assist.dla.mil - Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

(b) Launcher tube (Drawing 13-12-141).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Dimensions incorrect
102	Upper tube inside diameter incorrect
103	Lower base inside diameter incorrect
104	Tube wall thickness incorrect (upper and lower)
105	Base hole size incorrect
106	Base hole location incorrect
107	Surface finish incorrect
108	Color, surface, cracks and marking incorrect
109	Workmanship (3.5)

(c) Electrical connector (Drawing 13-12-146).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.065 percent defective	
101	Electrical short	4.4.4.2
102	Electrical non-continuity	4.4.4.2
Major:	AQL 1.0 percent defective	
103	Component missing or incorrectly assembled	
104	Dimensions incorrect	
105	Hole diameters incorrect	
106	Hole locations incorrect	
107	Wiring assembly length incorrect	
108	Marking missing or incorrect	
109	Electrical connections loose	
110	Welding incorrect	
111	Workmanship (3.5)	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

(d) Launcher platform (Drawing 13-12-138).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing or incorrectly assembled
102	Length incorrect
103	Width incorrect
104	Hole diameters incorrect
105	Hole locations incorrect
106	Bend radii incorrect
107	Pads not cemented adequately
108	Marking missing or incorrect
109	Finish incorrect or missing
110	Welding incorrect
111	Workmanship (3.5)

(e) Handle (Drawing 13-12-142).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Dimensions incorrect
102	Radii incorrect
103	Workmanship (3.5)

(f) Pad (Drawing 13-12-139 and 13-12-140).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Dimensions incorrect
102	Hole diameters incorrect
103	Hole locations incorrect
104	Workmanship (3.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

(g) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Preservation incorrect (5.1)
102	Packing incorrect (5.2)
103	Marking incorrect (5.3)

4.4.4 Tests. The tests shall be performed as follows:

4.4.4.1 Burst strength. The hydrostatic pressure of the launcher tube shall be conducted using test equipment approved by the Government to apply 450 psig for a minimum of 2 minutes and then increasing pressure gradually until the tube bursts. Record the burst pressure.

4.4.4.2 Continuity. The continuity shall be checked using an ohmmeter. The continuity between the igniter wires shall be tested when the test clips for the firing line are shorted.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Launcher platform (13-12-138), launcher tube (13-12-141), electrical connector (13-12-146), cable (50 meters), bolt (square neck round head 5 each) and manual (TM 3-1370-102-12). Each item shall be individually unit packed in accordance with Drawing 13-12-144.

5.1.2 Commercial.

5.1.2.1 Launcher platform (13-12-138), launcher tube (13-12-141), electrical connector (13-12-146), cable (50 meters), bolt (square neck round head 5 each) and manual (TM 3-1370-102-12). Each item shall be individually unit packed in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Launcher platform (13-12-138), launcher tube (13-12-141), electrical connector (13-12-146), cable (50 meters), bolts (square neck round head 5 each)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

and manual (TM 3-1370-102-12). Each item unit packed as specified in 5.1.1.1 shall be packed within the shipping container in accordance with Drawing 13-12-144.

5.2.2 Level B.

5.2.2.1 Launcher platform (13-12-138), launcher tube (13-12-141), electrical connector (13-12-146), cable (50 meters), bolts (square neck round head 5 each) and manual (TM 3-1370-102-12). Each item unit packed as specified in 5.1.1.1 shall be packed in accordance with Drawing 13-12-144, except that the shipping container shall be domestic type, style A, grade B of PPP-B-601.

5.2.3 Commercial.

5.2.3.1 Launcher platform (13-12-138), launcher tube (13-12-141), electrical connector (13-12-146), cable (50 meters), bolts (square neck round head 5 each) and manual (TM 3-1320-102-12). Each item unit packed as specified in 5.1.2.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required in the contract or order, unit and shipping containers shall be marked in accordance with Drawing 13-12-144 and ASTM D 3951, as applicable (see 6.2).

5.4 Repair parts. The launcher tube or electrical connector when procured as a repair part shall be preserved, unit packed, packed, and marked as specified in the Special Packaging Instructions (SPI) cited in the contract or order.

6. NOTES

6.1 Intended use. The M267 launcher is intended to be used for the M11 liquid airburst projectile for tactical training exercises.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation required (A or commercial) (see 5.1).
(c) Level of packing required (A, B, or commercial) (see 5.2).
(d) Marking required (see 5.3).
(e) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for Government to notify the contractor whether or not to proceed with production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

6.3 Supersession data. This specification supersedes and includes the requirements of Chemical Systems Laboratory Purchase Description EA-L-1265, dated 30 June 1983.

6.4 Subject term (key word) listing.

Launcher, projectile, liquid airburst
SPAL
Tube, launcher

Custodian

Army - EA

Preparing activity:

Army - EA

Project No. 6920-A072

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-51543(EA)	LAUNCHER, PROJECTILE, LIQUID AIRBURST: M267

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil/online/start/ downloaded: 2026-02-22T21:04Z
Check the source to verify that this is the current version before use.